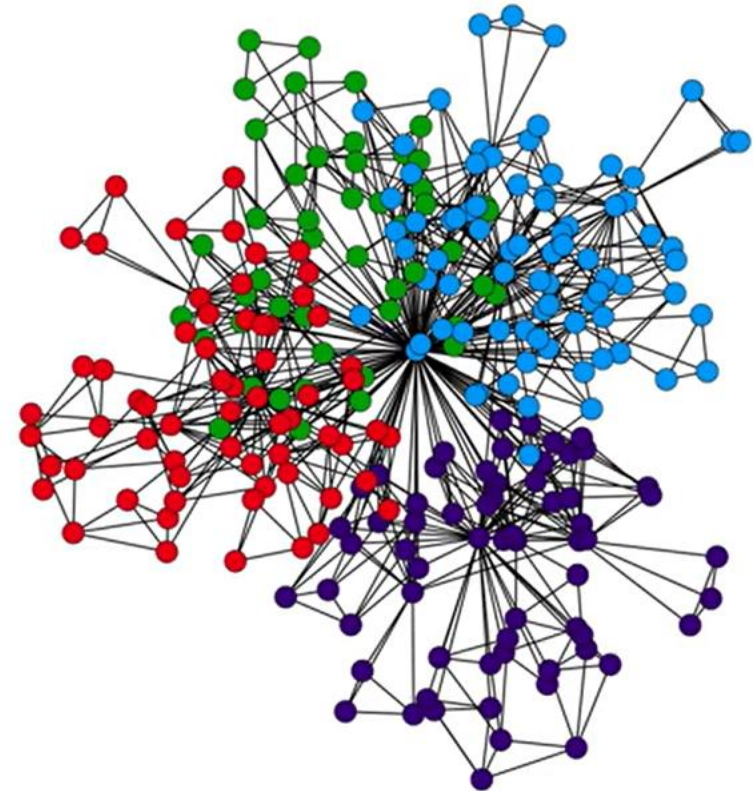


Networks

Networks: A set of objects (nodes) with interconnections (edges).

Why study networks?

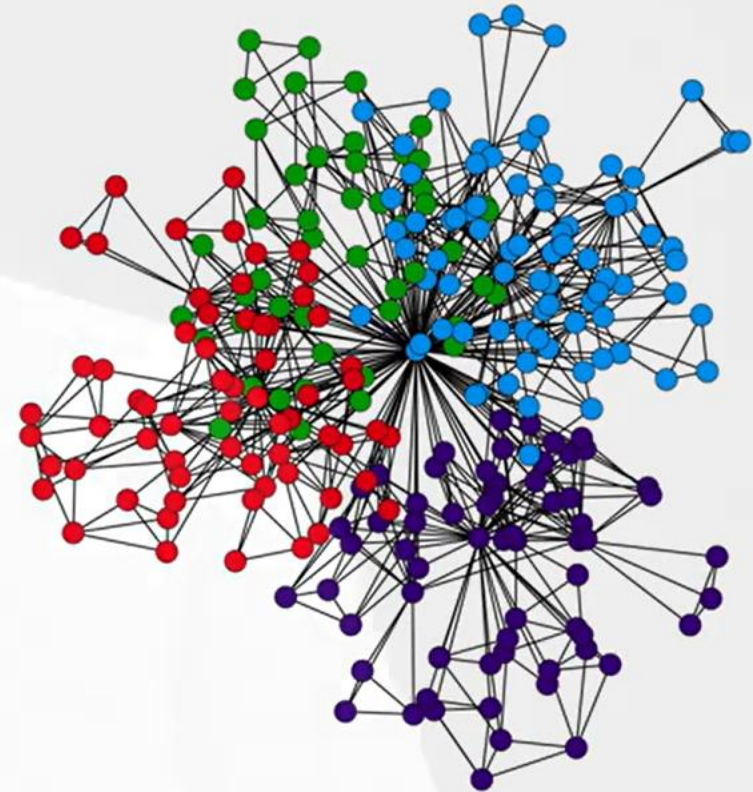


Networks

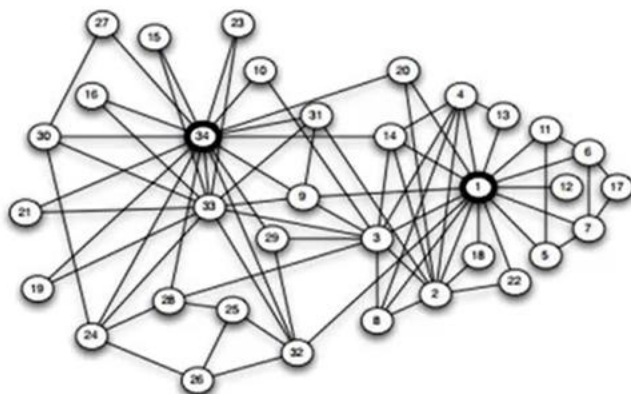
Networks: A set of objects (nodes) with interconnections (edges).

Why study networks?

Because they are everywhere!

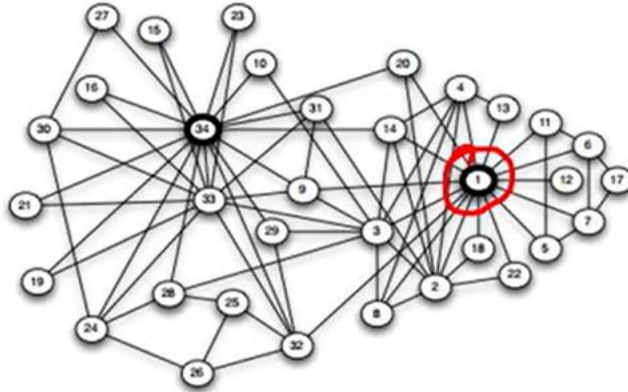


Social Networks

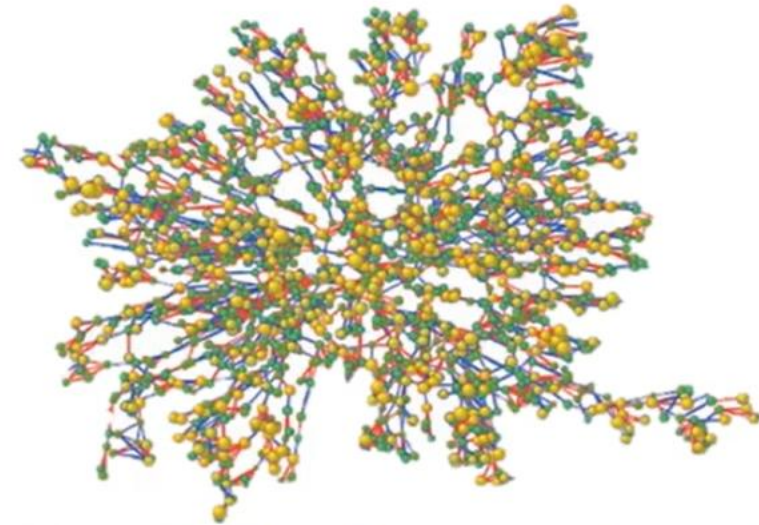


Friendship network in a 34-person karate club
[Zachary 1977]

Social Networks

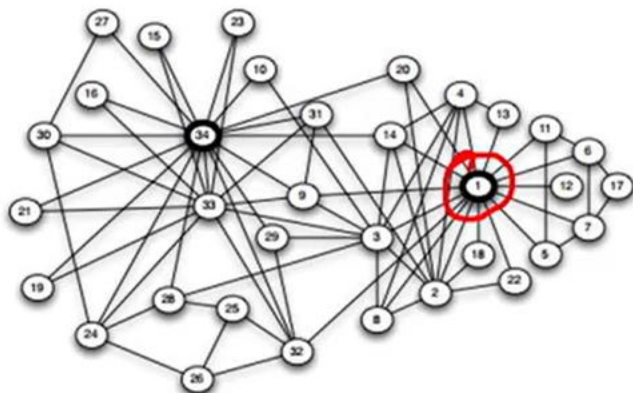


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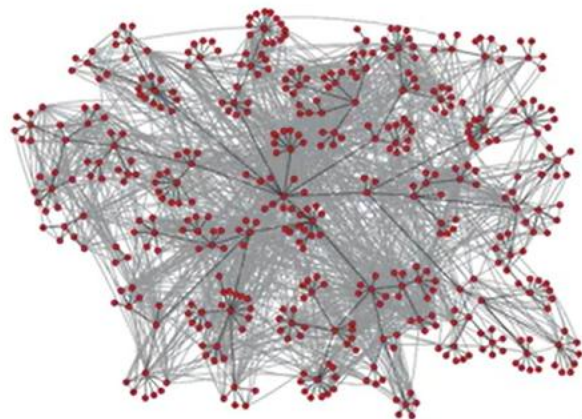


Network of friendship, marital tie, and
family tie among 2200 people
[Christakis & Fowler 2007]

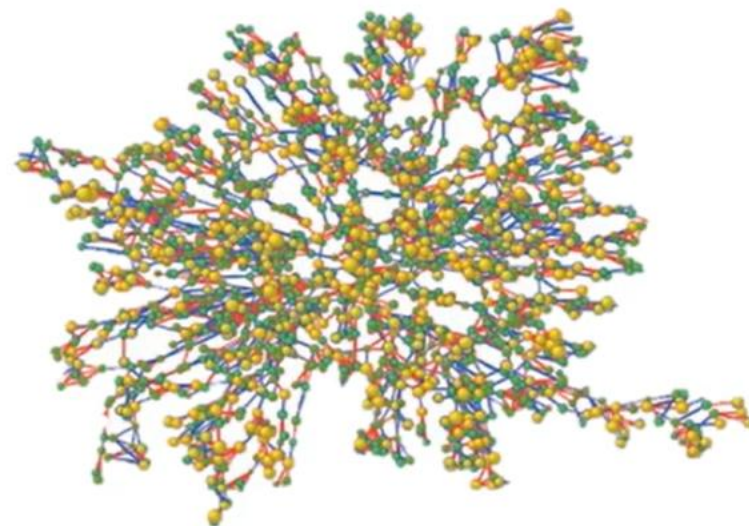
Social Networks



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E-mail communication network
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Transportation and Mobility Networks

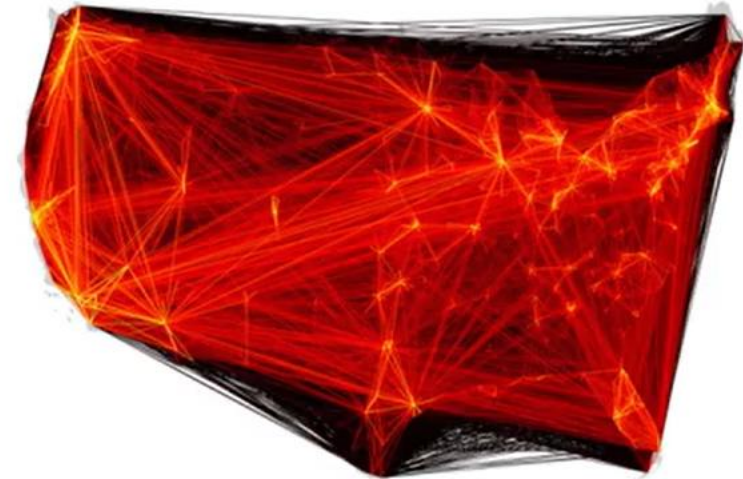


Network of direct flights around the world
[Bio.Diaspora]

Transportation and Mobility Networks



Network of direct flights around the world
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Human mobility network based
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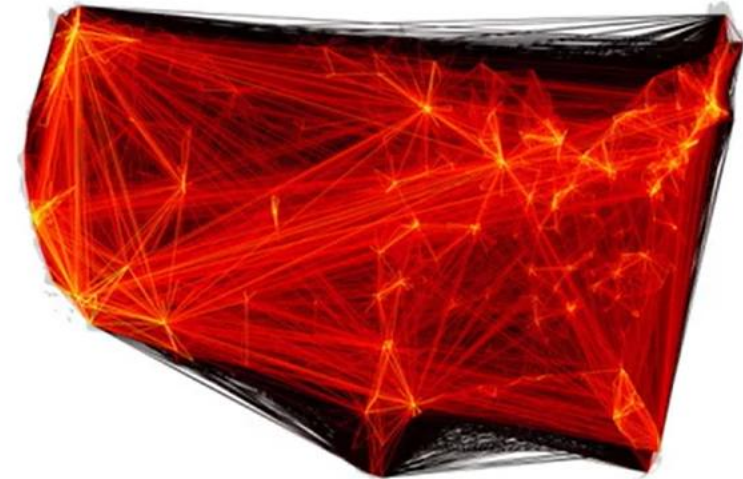
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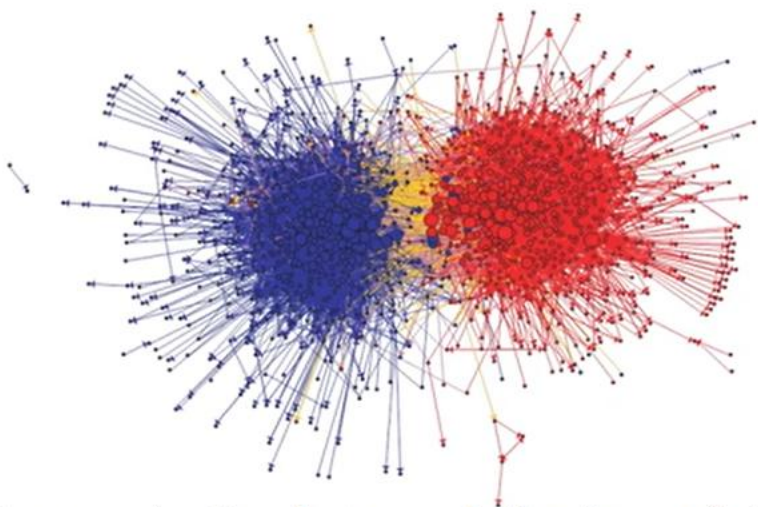


Ann Arbor bus transportation network



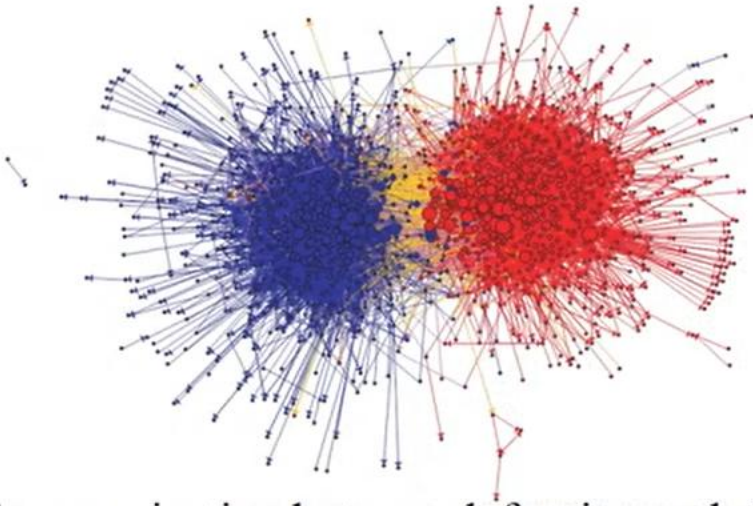
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Information Networks

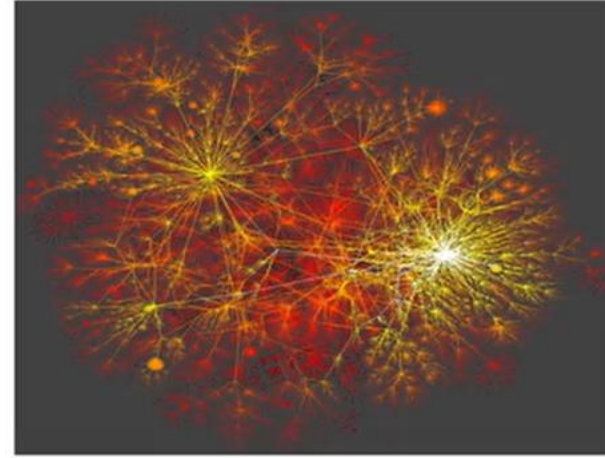


Communication between left-wing and right-wing political blogs [Adamic & Glance 2005]

Information Networks

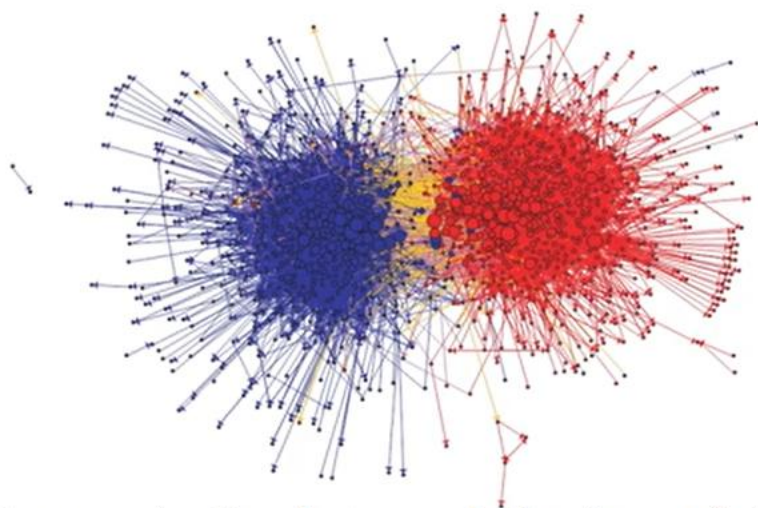


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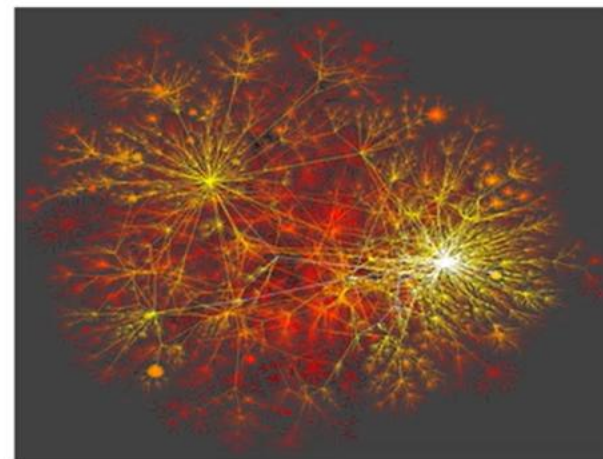


Internet Connectivity [K. C. Claffy]

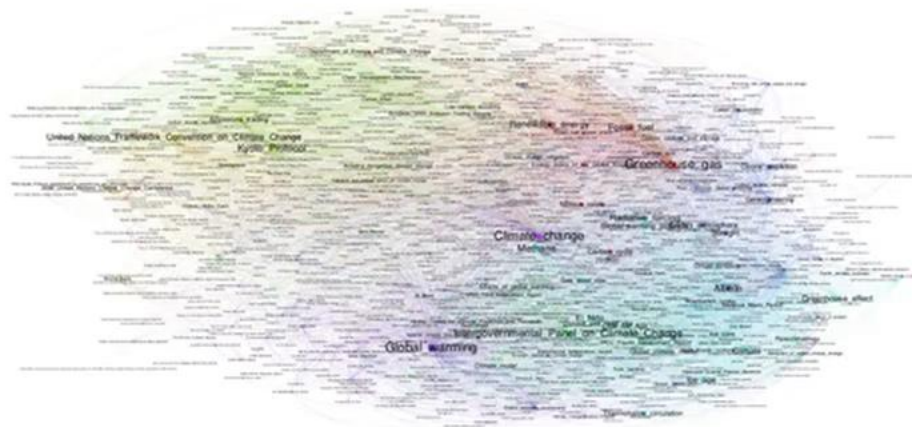
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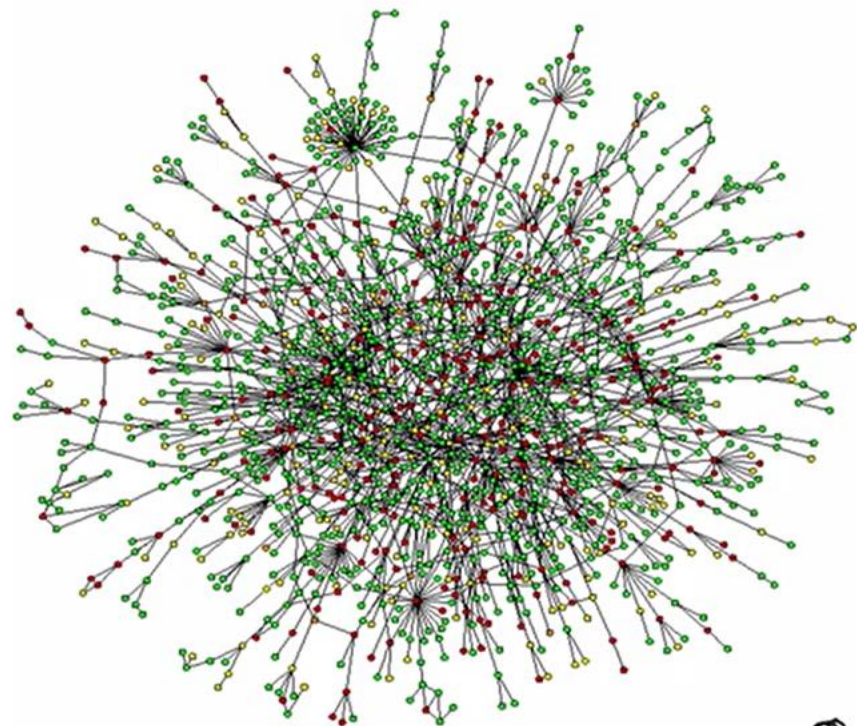


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Network of Wikipedia articles
about climate change [EMAPS]

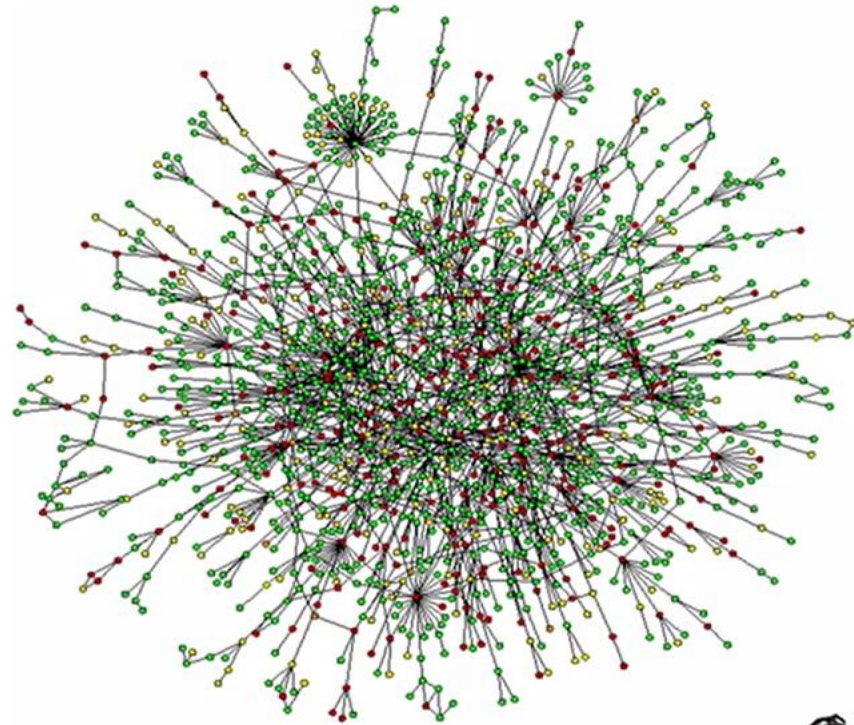
Biological Networks



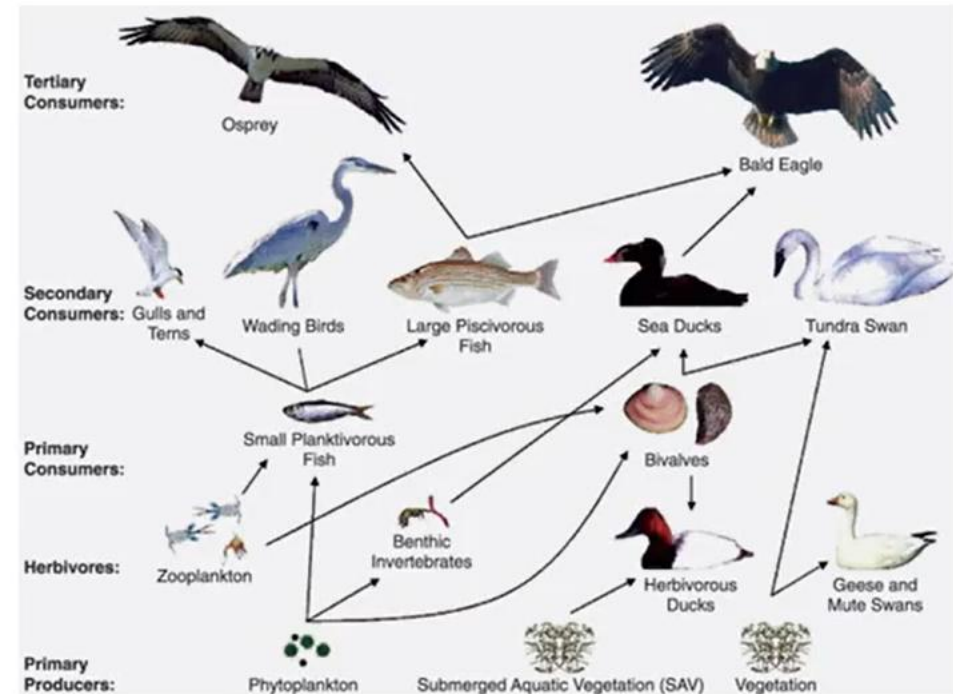
Protein-protein interactions
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Biological Networks



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Chesapeake Bay Waterbird Food Web
[Perry et al. 2005]

And More...

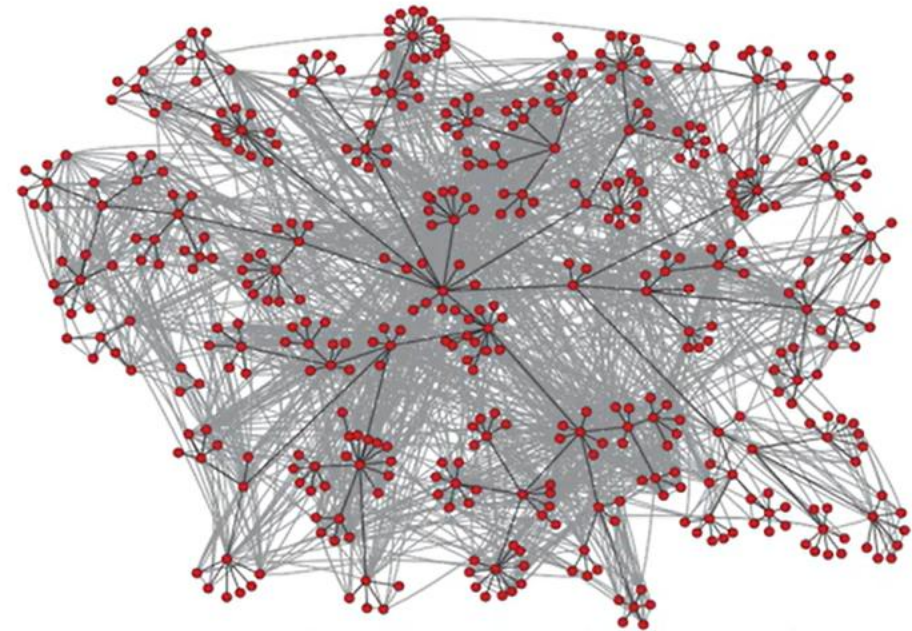
- Financial networks
- Co-authorship networks
- Trade networks
- Citation networks

Networks Applications

Networks are everywhere,
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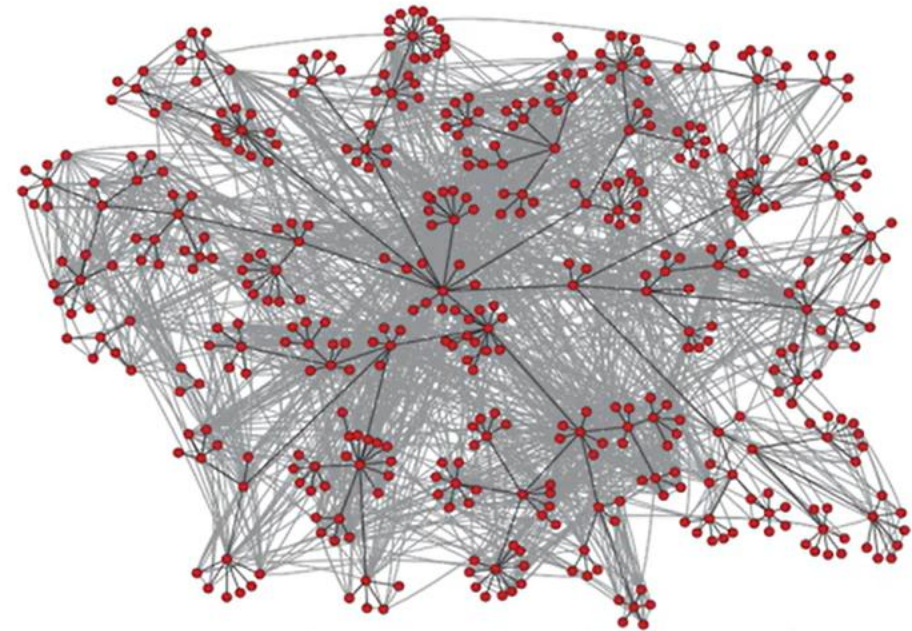
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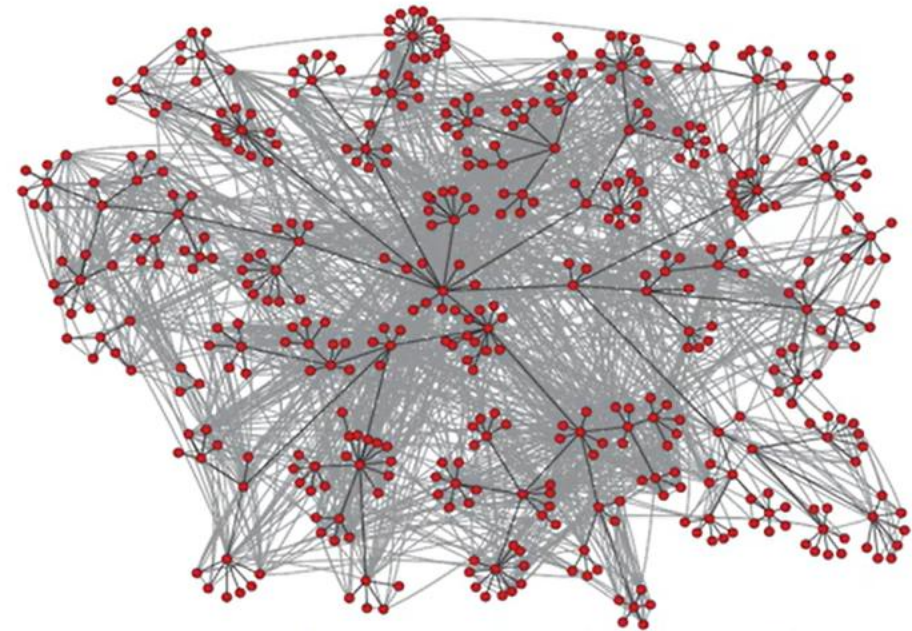
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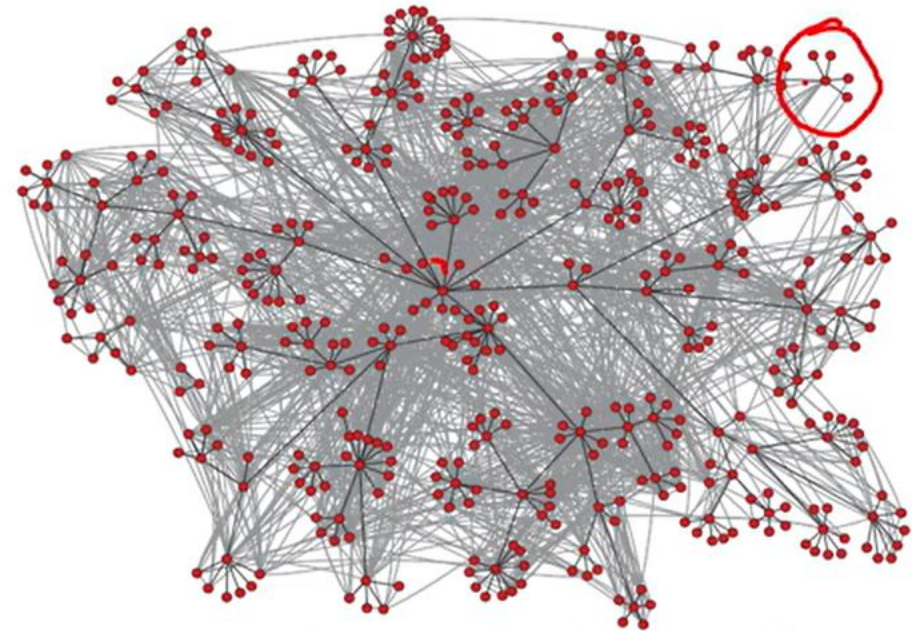


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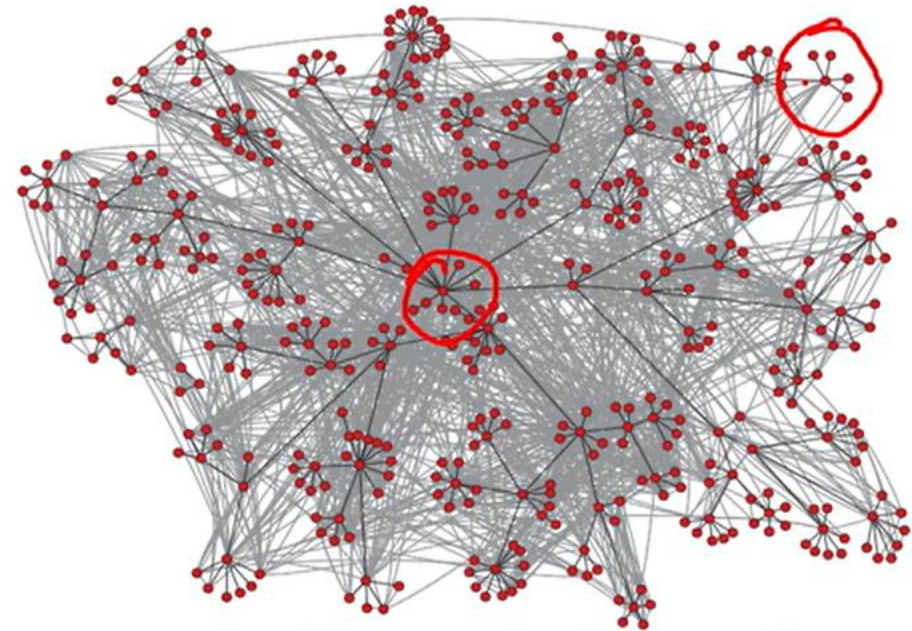


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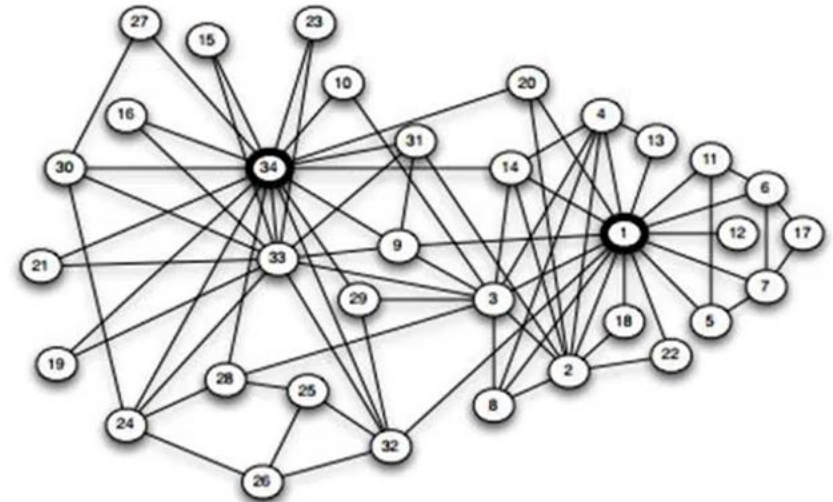


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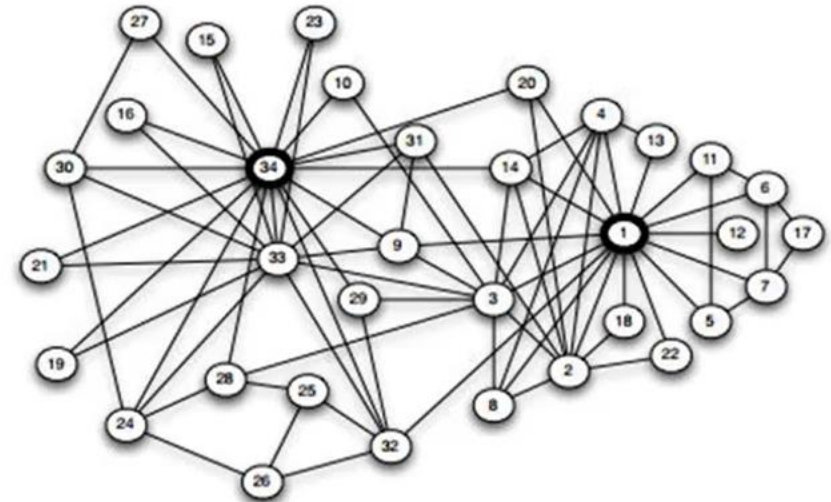


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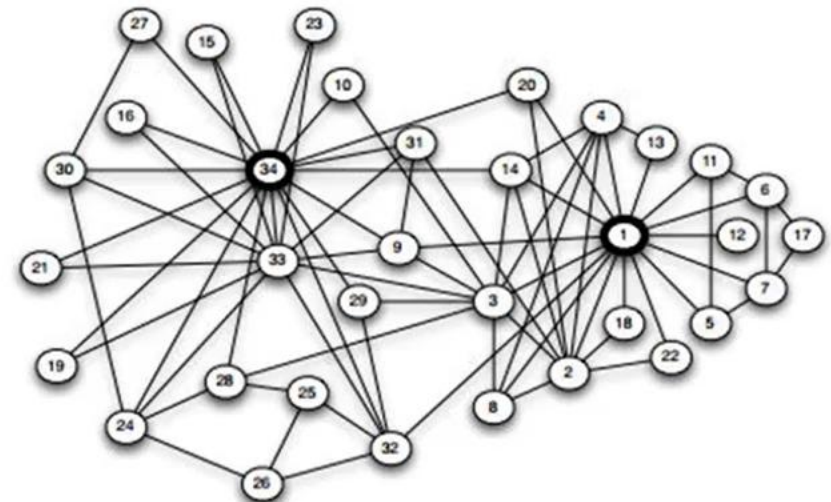
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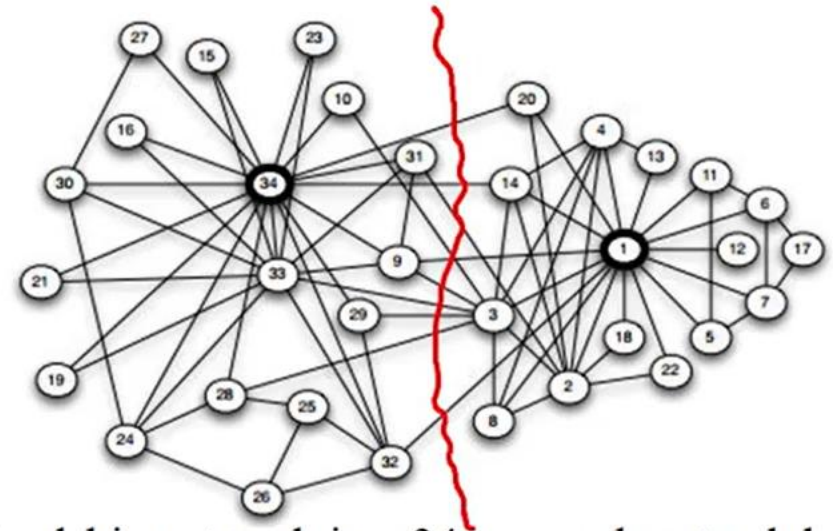
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Network of direct flights around the world

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Which airports are at highest risk for virus spreading?



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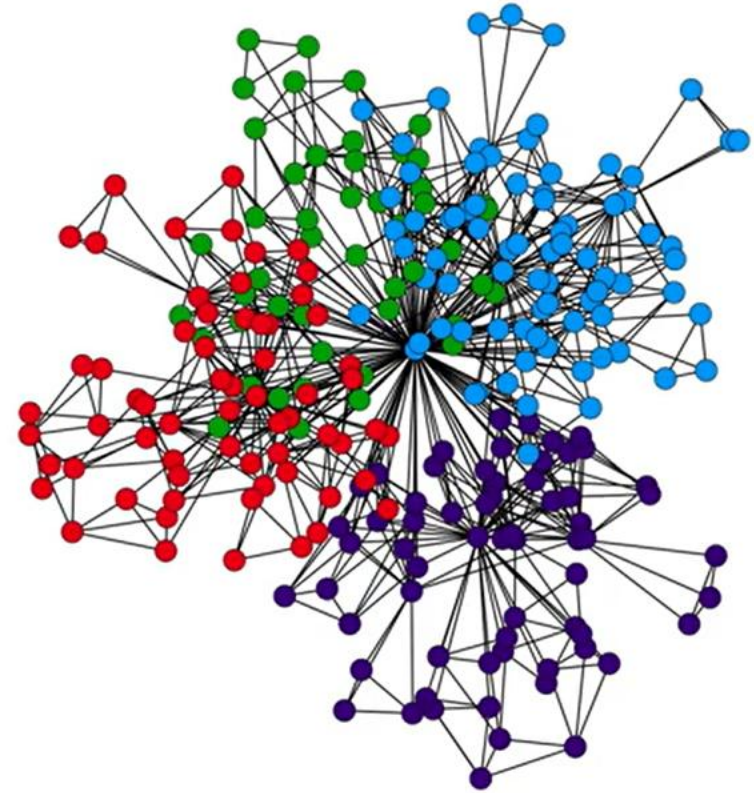
Are some parts of the world more difficult to reach?



Network of direct flights around the world

Summary

Many complex structures can be modeled by networks.



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Many complex structures can be modeled by networks.

Studying the structure of a network can allow us to answer questions about complex phenomena.

